

Lab 1

What is the difference between:

- **CMD & ENTRYPOINT**

- **CMD**

- **Function:** Sets a default command.
 - **Flexibility:** It is easily overridden if you provide a command at docker run.
 - **Example:** CMD ["python", "app.py"] (If you run docker run my-app bash, it ignores the python script).

- **ENTRYPOINT**

- **Function:** Sets a permanent command that makes the container act like a "Tool".
 - **Flexibility:** It is hard to override. Any arguments you provide at docker run are appended to it.
 - **Example:** ENTRYPOINT ["python"] (If you run docker run my-app script.py, it executes python script.py).

- **COPY & ADD**

- **COPY**

- The COPY instruction is straightforward and does exactly what its name implies: It copies files and directories from a source within the build context to a destination layer in the Docker image. This instruction can be used to copy both files and directories, and all paths on the host are relative to the root of the build context.
 - **Syntax**
 - <src>: The source files or directories on the host.
 - <dest>: The destination path inside the Docker image.

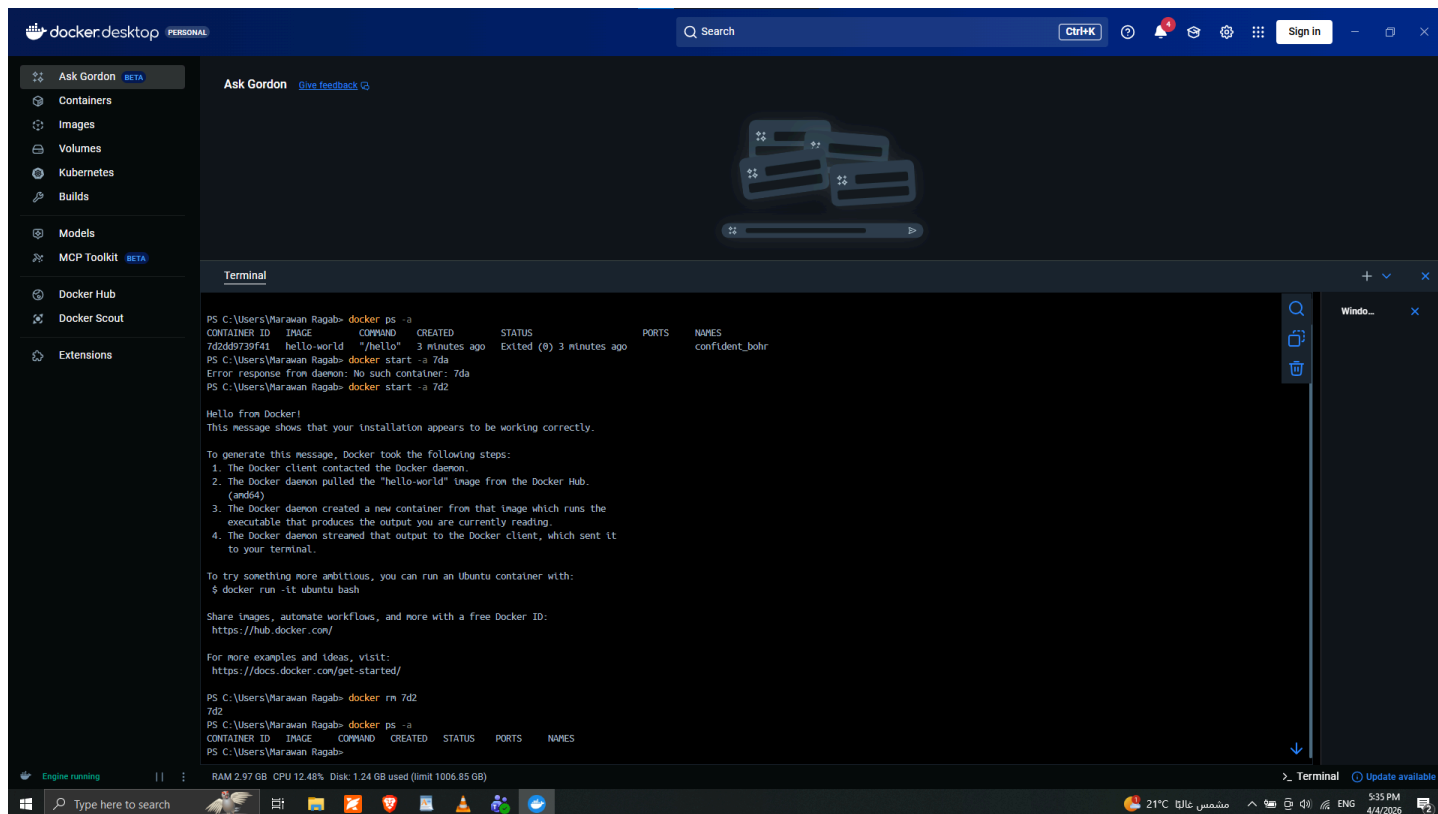
- **ADD**

- The ADD instruction provides the same functionality that the COPY instruction does, but it also has additional functionality that, if misunderstood, can introduce complexity and potential security risks.
 - **Syntax:**
 - <src>: The source files (directories or URLs).
 - <dest>: The destination path inside the Docker image.

Problem 1:

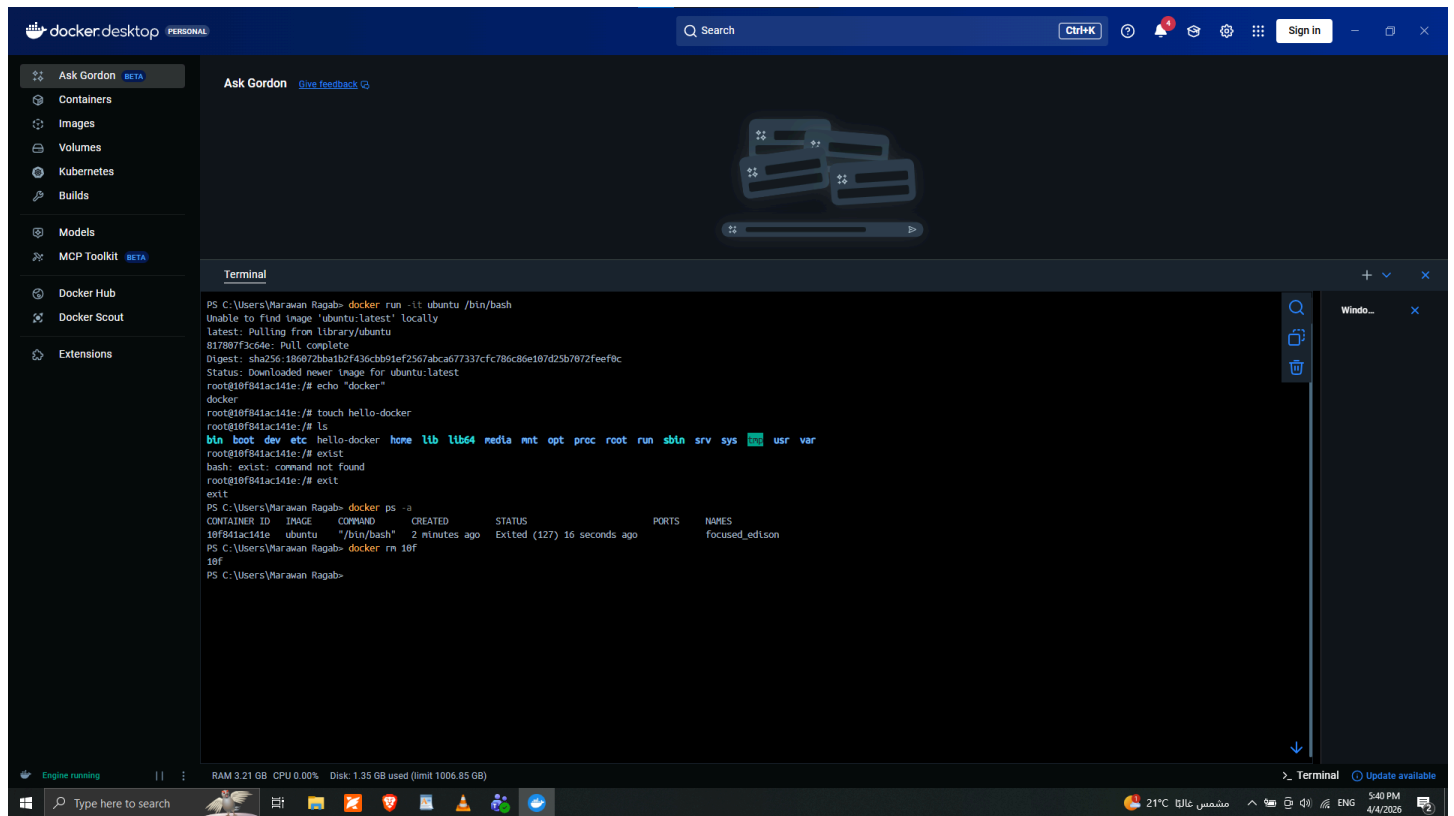
- **Run the container hello-world**

- Check the container status
- Start the stopped container
- Remove the container
- Remove the image



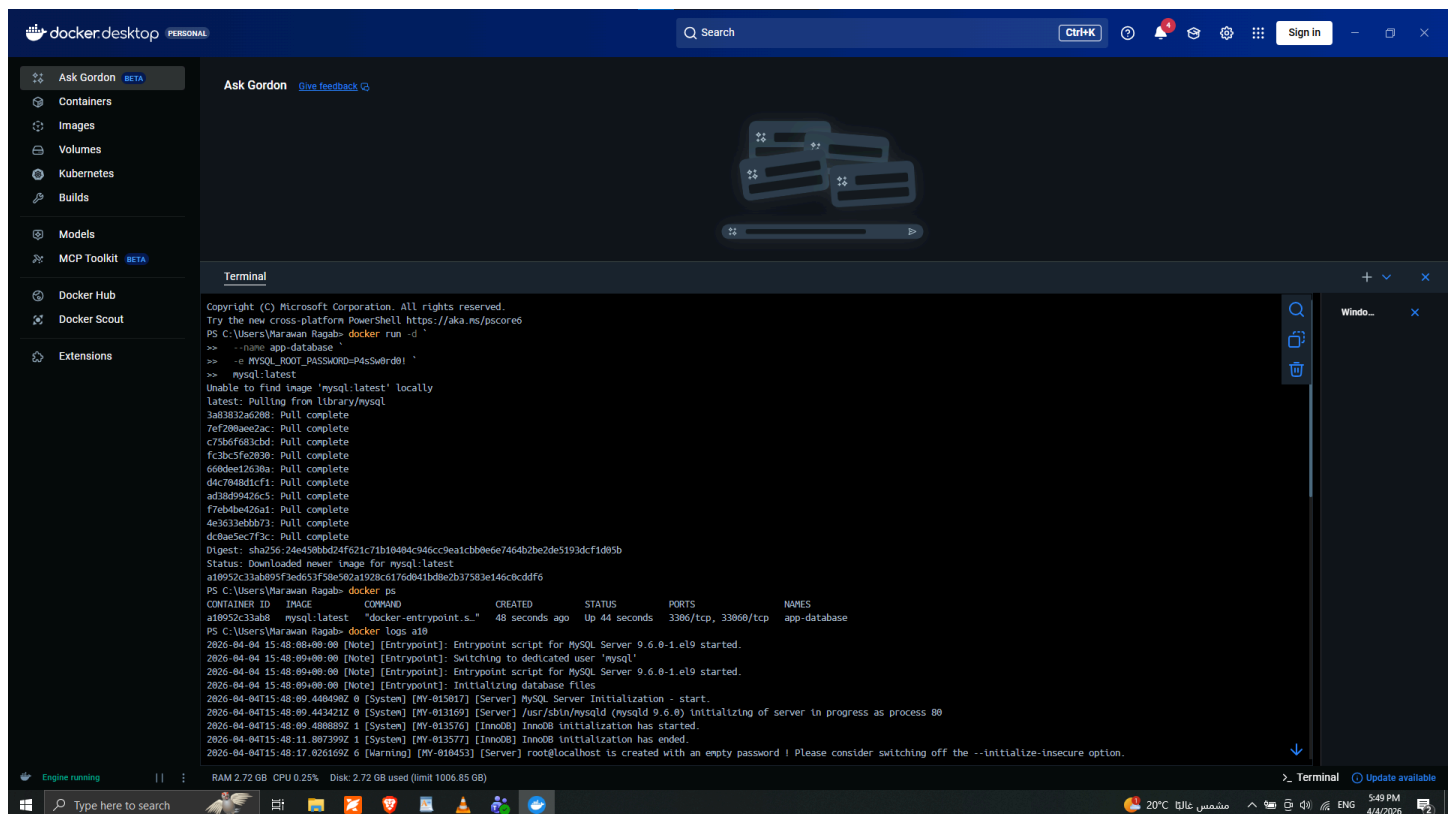
Problem 2:

- Run container centos or ubuntu in an interactive mode
- Run the following command in the container “echo docker ”
- Open a bash shell in the container and touch a file named hello-docker
- Stop the container and remove it. Write your comment about the file hello- docker
- Remove all stopped containers



Problem 3:

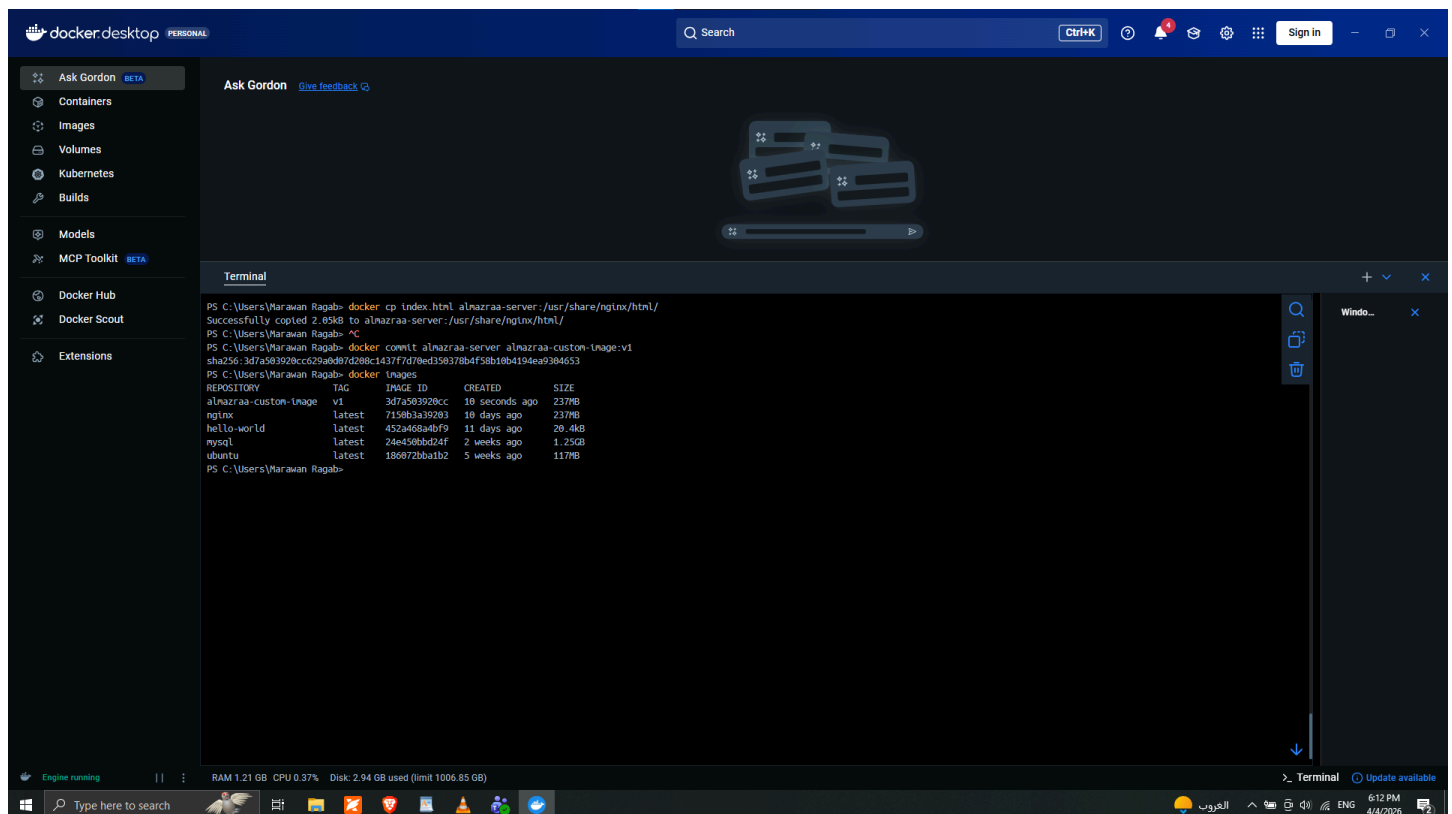
- Deploy a MySQL database called app-database. Use the mysql latest image, and use the -e flag to set MYSQL_ROOT_PASSWORD to P4sSw0rd0!. The container should run in the background.



Problem 4:

- Run the image Nginx
- Add html static files to the container and make sure they are accessible
- Commit the container with image name IMAGE_NAME

```
PS C:\Users\Marawan Ragab> docker run -d --name almazraa-server -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
510ddf6557d6: Pull complete
3189680c601f: Pull complete
587e3d84dbb5: Pull complete
5e815e07e569: Pull complete
bb3d0aa29654: Pull complete
cde7a05ae428: Pull complete
ec781dee3f47: Pull complete
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Downloaded newer image for nginx:latest
ea502abb8aaefa594ae361a3a80f0cee2d2f6085643c0e8523180323f84ff2e6
PS C:\Users\Marawan Ragab> echo "<h1>Welcome to Al Mazraa Project </h1>" > index.html
```



Problem 5:

- Create a python simple app
- Create a dockerfile to containerize the python app
- Build the image and test it
- (Bonus)create a dockerfile for the same app in smaller size using multi staging

▪ Push the created image into your docker hub repo

```
test.py x DockerFile 1
test.py
1 print("Hello world")
```

The screenshot shows the Visual Studio Code interface with a Dockerfile open in the editor and its build output in the terminal.

Dockerfile:

```
1 From python:3.9-slim
2
3 WORKDIR /app
4
5 COPY test.py .
6
7 CMD ["python", "test.py"]
```

Terminal Output:

```
F:\ITI 9 Months AI\Labs\15. Docker\Lab 01\ren Dockerfile Dockerfile
f) Building 37.0s (0/0) FINISHED
-> [internal] load build definition from Dockerfile
-> transferring dockerfile: 123B
-> WARN: ConsistentInstructionCasing: Command 'From' should match the case of the command majority (uppercase) (line 1)
-> [internal] load metadata for docker.io/library/python:3.9-slim
-> [internal] load dockerignore
-> transferring context: 2B
-> [1/3] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aabb1acda1e13cf1731b1b
-> resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aabb1acda1e13cf1731b1b
-> sha256:ea56f6854b4ad81688322f152d2cfc62115b3bdaa81c2c450878315eb508 251B / 251B
-> sha256:fc74c3804922d130a44b2069a953f842f59c6d11a8c2c3d710fffa286c08 13.89MB / 13.89MB
-> sha256:b3ec39b36ae8c83a3e90854d4ec4aa8381dfed84a0daa07504a2c03df3881d 1.29MB / 1.29MB
-> sha256:38513bd7256313495cdd83b3b0915a633cf475dc2a0702ab2c8d191020ca5d 29.78MB / 29.78MB
-> extracting sha256:38513bd7256313495cdd83b3b0915a633cf475dc2a0702ab2c8d191020ca5d
-> extracting sha256:b3ec39b36ae8c83a3e90854d4ec4aa8381dfed84a0daa07504a2c03df3881d
-> extracting sha256:fc74c3804922d130a44b2069a953f842f59c6d11a8c2c3d710fffa286c08
-> extracting sha256:ea56f6854b4ad81688322f152d2cfc62115b3bdaa81c2c450878315eb508
-> [internal] load build context
-> transferring context: 54B
-> [2/3] WORKDIR /app
-> [3/3] COPY test.py .
-> exporting to image
-> exporting layers
-> exporting manifest sha256:f641edd295547b699d366e5b7a22443258a0e7fa851a2466735046e4f7b4b8
-> exporting config sha256:68944026a33fb4f0cc3b00eac18070cf6023399e17b7a60f1c5b6e98f19e
-> exporting attestation manifest sha256:9c8038c538556445c2d2180234e441101dc84e1e61402fc918d798297475c
-> exporting manifest list sha256:7b6101d67c188423990b40cd75816f6e00b72a31a173b83c053c530952ad891
-> naming to docker.io/library/test:latest
-> unpacking to docker.io/library/test:latest
```

The screenshot shows the Visual Studio Code interface with a Dockerfile open in the editor and its build output in the terminal. The Dockerfile defines a multi-stage build for a Python application. The terminal shows the build process for 'test:v2', including downloading base images, copying source files, and exporting the final image.

```
Dockerfile
1 FROM python:3.9-slim as bu
2
3 WORKDIR /build
4
5 COPY test.py .
6
7
8 FROM python:3.9-alpine
9
10 WORKDIR /app
11 COPY --from=bu /build/test.py .
12
13 CMD ["python", "test.py"]
```

```
F:\ITI 9 Months Ai\Labs\15. Docker\Lab 01>docker build -t test:v2 .
[*] Building 29.0s (12/12) FINISHED
-> [internal] load build definition from Dockerfile
-> > transferring dockerfile: 210B
-> WARN: FromAsCasing: 'as' and 'FROM' keywords' casing do not match (line 1)
-> [internal] load metadata for docker.io/library/python:3.9-alpine
-> [internal] load metadata for docker.io/library/python:3.9-slim
-> [internal] load .dockerignore
-> > transferring context: 2B
-> CACHED [bu 1/3] FROM docker.io/library/python:3.9-slim@sha256:2d97f6918b16bd338d38d3860f261f53f144965f755599aab1acda1e13cf1731b1b
-> > resolve docker.io/library/python:3.9-slim@sha256:2d97f6918b16bd338d38d3860f261f53f144965f755599aab1acda1e13cf1731b1b
-> [internal] load build context
-> > transferring context: 28B
-> [stage-1 1/3] FROM docker.io/library/python:3.9-alpine@sha256:c99b6eb43b3ac4d750db3d6e8b22268d5ea9a99deead7218ce3deda7f2ca029c
-> > resolve docker.io/library/python:3.9-alpine@sha256:c99b6eb43b3ac4d750db3d6e8b22268d5ea9a99deead7218ce3deda7f2ca029c
-> > sha256:980628fa0da943d2440c4243bd7caa253eef05e8669881cee924ceecabe3b3 247B / 247B
-> > sha256:27711589a588c076752866b7f5d51377b4db350cf1f9a9213d0c3b08ed 15.40MB / 15.40MB
-> > sha256:c908197b8d9d9b4c7a87e38fa6dc8bceb4f7d87df8e9061284d123459f9451 456.92kB / 456.92kB
-> > sha256:2d35ebdb57d9971fe0cac1582aa78935adf805802c32db163c98822e5dfa1b 3.89MB / 3.89MB
-> > extracting sha256:2d35ebdb57d9971fe0cac1582aa78935adf805802c32db163c98822e5dfa1b
-> > extracting sha256:c908197b8d9d9b4c7a87e38fa6dc8bceb4f7d87df8e9061284d123459f9451
-> > extracting sha256:27711589a588c076752866b7f5d51377b4db350cf1f9a9213d0c3b08ed
-> > extracting sha256:980628fa0da943d2440c4243bd7caa253eef05e8669881cee924ceecabe3b3
-> [bu 2/3] WORKDIR /build
-> [bu 3/3] COPY test.py .
-> [stage-1 2/3] WORKDIR /app
-> [stage-1 3/3] COPY --from=bu /build/test.py .
-> exporting to image
-> exporting layers
-> exporting manifest sha256:c257ac8333d27bfc46b20894cbfc5e4a00ac9618a3dedc13d3d934561604873
-> exporting config sha256:af4f2188385e6cc3f09aaef0c09aa1105a27f3d88187704d9c4aa2d8981fae1
-> exporting attestation manifest sha256:385ef5f68f0385a04e37670a0762739803c480430b1dadff690cedf4e910da
-> exporting manifest list sha256:90c245cc98bbe118c99aa95ae2f831537073f0c70285f2285d6e61f4cb3992a
```

```
F:\ITI 9 Months Ai\Labs\15. Docker\Lab 01>docker run test
Hello world
```

```
F:\ITI 9 Months Ai\Labs\15. Docker\Lab 01>docker push marwan311/test:v2
The push refers to repository [docker.io/marwan311/test]
22b60225c55e: Pushed
18a1ce34b607: Pushed
b3ec39b36ae8: Pushed
38513bd72563: Pushed
5ac7f65c2073: Pushed
ea56f685404a: Pushed
fc7443084902: Pushed
v2: digest: sha256:7b6191d67c1d88423996bf0cd75816fe606bf2a31a173b83c653c530952ad891 size: 856
```

hub

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Repositories

All repositories within the marwan311 namespace.

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All content

Create a repository

Name	Last Pushed	Contains	Visibility	Scout
marwan311/test	10 minutes ago	IMAGE	Public	Inactive

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Upgrade plan